

Beong-woo Kwak

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LLM researcher specializing in agent post-training and scalable evaluation pipelines, with 12+ publications at NeurIPS/ICLR/ACL/EMNLP/AAAI and industry experience at Amazon and Microsoft Research.

SELECTED PUBLICATIONS

12+ papers at top-tier AI/NLP venues (NeurIPS, ICLR, EMNLP, ACL, NAACL, AAAI, 2022–2026)

Memento (*ICLR 2026*) – Memory utilization framework enabling personalized assistance in embodied agents.

ToolHaystack (*EMNLP 2025 Findings*) – Revealed systematic robustness failures in state-of-the-art tool-augmented LLMs under noisy long-term interactions.

Web-Shepherd (*NeurIPS 2025*) – Domain-agnostic process reward models for reinforcing web agents; scalable reward signal for multi-step web navigation.

Coffee-Gym (*EMNLP 2024*) – Environment for evaluating and improving natural language feedback on erroneous code; enables RL-based training for code correction pipelines.

TrustAL (*AAAI 2022*) – Trustworthy active learning pipeline using knowledge distillation for mitigating catastrophic forgetting in low-resource model training.

EXPERIENCE

Microsoft Research Jun. 2025 – Dec. 2025

Research Scientist Intern (General Intelligence Group)

Beijing, China

- **Project 1 (Agentic RL):** Identified horizon length itself as a fundamental RL training bottleneck in LLM agents; proposed *horizon reduction* (macro-actions, subgoal decomposition) achieves 91% pass@4 and generalizes to unseen longer horizons (61% pass@4). (under review, incl. *ICML 2026*)
- **Project 2 (Self-evolving Agents):** Built a hypothesis-verification framework for self-evolving LLM agents that prevents memory mis-evolution based on empirical validation; evaluated on the AppWorld benchmark (multi-app, multi-step agent tasks).

Amazon (*Return Intern*)

Jun. 2024 – Sep. 2024

Applied Scientist Intern (AGI Foundational Models Group)

Sunnyvale, CA, USA

- Designed a difficulty-conditioned CoT-distillation pipeline for improving automated test case generation in Code LLMs; extended with RL using generated test cases as process reward signals.
- Improved test case success rate and line coverage across **3 internal Amazon models**; eval framework, corpus pipeline, and training infrastructure deployed internally.

Amazon

Jun. 2023 – Sep. 2023

Applied Scientist Intern (Alexa AI)

Sunnyvale, CA, USA

- Built a benchmark exposing that post-deployment tool updates (schema changes, new tools) cause significant robustness failures in SFT-trained LLMs, which is a previously unmeasured reliability gap in production tool-augmented agents.
- Developed a training method using symbolic tool representations with an auxiliary shortcut-debiasing task objective, achieving **+4% on seen tools** and **+13% robustness improvement** on post-update tool scenarios over baseline. This involved SFT training of 8B- and 13B-parameter models.

TECHNICAL SKILLS

Languages: Python, Java

ML Frameworks: PyTorch, Transformers, vLLM, LLaMA Factory, VeRL, Megatron, DeepSpeed

Infrastructure: AWS, Azure, Docker, Git, Distributed Training (LLM training with 150+ A100s)

Agent Tools: LangChain, Ollama

Research Areas: Agentic AI (w/ Tool Use & Memory), Agent Evaluation & Benchmarking, Post-training, Personalization

EDUCATION

Yonsei University

Seoul, Korea

Ph.D. Student, Artificial Intelligence (Advisor: Prof. Jinyoung Yeo) | GPA: 4.15/4.5

Mar. 2022 – Present

Yonsei University

Seoul, Korea

M.S., Artificial Intelligence | GPA: 4.20/4.5

Mar. 2020 – Feb. 2022

Yonsei University

Seoul, Korea

B.S., Computer Science | Major GPA: 3.86/4.5

Mar. 2014 – Feb. 2022

HONORS & AWARDS

Encouragement Prize – Korea Capstone Design Fair, Ministry of Trade, Industry and Energy (Yonsei Univ. representative)

Top Prize – Software Capstone Design, Dept. of Computer Science, Yonsei Univ. (*selected among 30+ teams*)

Top Presentation Prize – Yonsei Creative Exhibition, College of Engineering, Yonsei Univ.